

# VIGNESH KUMAR EKAMBARAM

## TECH LEAD

Full-Stack Architecture | Client Delivery | Team Leadership

Chennai, Tamil Nadu, India | 12+ years in software development | Open to international and on-site assignments

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## PROFESSIONAL PROFILE

Tech Lead with 12+ years of experience delivering distributed, high-performance enterprise software. Leads a five-member engineering team, manages client priorities and delivery coordination, and combines hands-on expertise in .NET Core backend development, Angular, React, SQL Server, AWS, automation, machine learning, and AI/RAG initiatives. Has supported client engagements through multiple on-site visits to Thailand and a three-month client visit to Japan to gather product-usage feedback.

## LEADERSHIP & DELIVERY STRENGTHS

- **Team leadership:** Lead a five-member engineering team; plan work, unblock delivery, coordinate stakeholders, and manage client requests.
- **International client support:** Supported client visits and delivery activities in Thailand and Japan.
- **Quality engineering:** Improved browser-level regression coverage using Playwright automation and MCP-enabled testing workflows.
- **Cloud delivery:** Experience with AWS EC2, S3, ELB/load balancing, CloudSearch, WAF, and RDS for SQL Server and MySQL.

## TECHNICAL SKILLS

|                                                                                                                                                           |                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages &amp; Frameworks</b><br>C#, TypeScript, JavaScript, Python, SQL, .NET Core, Web API, EF Core, WCF, Angular, React, Node.js, Express.js, YARP | <b>Cloud &amp; DevOps</b><br>AWS EC2, S3, ELB/ALB, CloudSearch, RDS, WAF, Docker, GitLab CI/CD, Firebase, Vercel, IIS, Nginx, Apache, Linux    |
| <b>Data &amp; Distributed Systems</b><br>SQL Server, MySQL, Oracle, Query Store, Redis, RabbitMQ, AWS SQS, SignalR, WebSockets                            | <b>AI, ML &amp; Quality</b><br>Qdrant, RAG, AI Agents, MCP, HOG, YOLO, scikit-learn, DecisionTreeRegressor, Playwright, SOLID, design patterns |

## PROFESSIONAL EXPERIENCE

### Tech Lead

alfaTKG  
2022 - Present

- Lead the engineering delivery of JQMS, PTE, and AlfaDock using .NET Core for backend development, alongside Angular, React, Node.js, and SQL Server.
- Work directly with clients, manage project priorities and delivery, and lead a five-member engineering team.
- Provide on-site client support in Thailand and Japan, gather requirements and product feedback, and coordinate issue resolution.
- Designed asynchronous processing with RabbitMQ, AWS SQS, and .NET Core worker services; used SignalR/WebSockets for live feedback.

- Optimized high-concurrency SQL Server workloads through Query Store analysis, execution plans, locking analysis, and indexing.
- Architected AWS deployments and automated GitLab CI/CD pipelines for multi-tenant IIS environments.

## Senior Software Engineer / Full-Stack Engineer

alfaTKG  
2018 - 2022

- Gathered requirements, prepared specifications, planned designs, managed cloud servers and databases, and delegated team and project tasks.
- Developed REST APIs and full-stack applications; managed AWS EC2, RDS, CloudSearch, and load balancing.
- **GAIA Quote and alfaDOCK.** Built quotation and cloud document workflows for small manufacturers. Developed a TypeScript/Node.js/Express quotation API using 3D JSON data from S3, a MySQL-backed report scheduler, and an IIS-hosted C# thumbnail service storing outputs in S3.
- **GAIA Smart Quote (GSQ).** Collected customer sales-order data, filtered and preprocessed inputs and outputs, trained Python/scikit-learn regression models, evaluated them using mean squared error (MSE), and developed the Angular frontend.
- Applied SOLID principles and design patterns; worked with reflection-based plugins, dependency injection, middleware, validation, logging, Swagger, Angular lazy loading, and RxJS in Agile delivery.
- Built machine-learning workflows for sheet-metal quotation and machine-cycle-time prediction.
- Applied HOG feature extraction, YOLO-based visual detection, and regression approaches including DecisionTreeRegressor to manufacturing data.
- Designed modular quotation engines for laser cutting, punching, bending, welding, cost estimation, and BOM generation.
- Developed core architecture for AlfaDock and PTE, and mentored engineers on testing and design practices.

## Software Development Engineer

alfaTKG  
Aug 2015 - 2018

- Developed Angular frontend applications and .NET desktop applications using WPF; supported C++ wrapper libraries for integration into C# applications.
- Developed and maintained applications and REST APIs; fixed bugs, provided support, managed Git workflows and server deployments, and reported daily progress.

## Software Engineer

Sirpi  
Oct 2013 - Aug 2015

- Developed Windows desktop applications using .NET Framework and worked on Android app development; prepared test cases and performed testing.

## SELECTED ENGINEERING HIGHLIGHTS

**Manufacturing quotation intelligence.** Automated sheet-metal quotation workflows with CAD-driven inputs, nesting calculations, laser-cutting paths, bending operations, and machine-runtime estimation.

**AI agents and RAG.** Designed specialized quotation, production scheduling, and machine-data agents coordinated through an orchestrator using Qdrant retrieval and LLM reasoning.

**Resilient enterprise systems.** Implemented YARP API gateway patterns, JWT authentication, Redis caching, rate limiting, and decoupled worker-based processing.

## EDUCATION

**B.E. in Electronics and Communication Engineering** | 2009 - 2013  
Hidusthan College, Coimbatore, Anna University | GPA: 8.4